PRACTICAL-13

Create simple PHP JSON API endpoints for events or students and consume them using Fetch API. Perform list, add, update, and delete operations asynchronously without full page reload.

Code:

Database

```php
<?php
$conn = mysqli_connect("localhost", "root", "", "StudentHub");

if (!$conn) {
    die("Database connection failed");
}
?>
```

Registration.php

```
<!DOCTYPE html>
<html>
<head>
    <title>Student Registration</title>

    <style>
        body {
            font-family: Arial;
            margin: 30px;
            background: #f2f2f2;
        }

        input {
            padding: 8px;
            margin: 5px;
        }

        button {
            padding: 8px;
            margin: 3px;
            cursor: pointer;
        }

        table {
```

```
            background: white;
            margin-top: 20px;
        }

        th, td {
            padding: 10px;
            border: 1px solid black;
        }
    </style>
</head>

<body>

<h2>Student Registration Form</h2>

<form id="myForm">
    <input type="text" id="name" placeholder="Name" required>
    <input type="email" id="email" placeholder="Email" required>
    <input type="text" id="mobile" placeholder="Mobile" required>

    <button type="submit">Register</button>
</form>

<p id="msg"></p>

<h3>Student List</h3>

<table border="1">
    <thead>
        <tr>
            <th>ID</th>
            <th>Name</th>
            <th>Email</th>
            <th>Mobile</th>
            <th>Action</th>
        </tr>
    </thead>

    <tbody id="list"></tbody>
</table>

<script>
const api = "api/students.php";
```

```
// LIST
function showStudents() {
    fetch(api)
    .then(res => res.json())
    .then(data => {
        let rows = "";

        data.forEach(s => {
            rows += `<tr>
                <td>${s.id}</td>
                <td>${s.name}</td>
                <td>${s.email}</td>
                <td>${s.mobile}</td>
                <td>
                      <button onclick="editStudent(${s.id}, '${s.name}',
'${s.email}', '${s.mobile}')">Update</button>
                                                                    <button
onclick="deleteStudent(${s.id})">Delete</button>
                </td>
            </tr>`;
        });

        document.getElementById("list").innerHTML = rows;
    });
}

// ADD
document.getElementById("myForm").onsubmit = function(e) {
    e.preventDefault();

    let student = {
        name: document.getElementById("name").value,
        email: document.getElementById("email").value,
        mobile: document.getElementById("mobile").value
    };

    fetch(api, {
        method: "POST",
        body: JSON.stringify(student)
    })
    .then(res => res.json())
    .then(data => {
```

```
        document.getElementById("msg").innerText = data.message;
        document.getElementById("myForm").reset();
        showStudents();
    });
};

// UPDATE
function editStudent(id, name, email, mobile) {
    let newName = prompt("Enter name", name);
    let newEmail = prompt("Enter email", email);
    let newMobile = prompt("Enter mobile", mobile);

     if (newName === null || newEmail === null || newMobile === null)
return;

    fetch(api, {
        method: "PUT",
        body: JSON.stringify({
            id: id,
            name: newName,
            email: newEmail,
            mobile: newMobile
        })
    })
    .then(res => res.json())
    .then(data => {
        document.getElementById("msg").innerText = data.message;
        showStudents();
    });
}

// DELETE
function deleteStudent(id) {
    if (confirm("Delete this student?")) {
        fetch(api + "?id=" + id, {
            method: "DELETE"
        })
        .then(res => res.json())
        .then(data => {
            document.getElementById("msg").innerText = data.message;
            showStudents();
        });
    }
```

```
}

showStudents();
</script>

</body>
</html>
```

Student.php

```
<?php
header("Content-Type: application/json");
include "../db.php";

$method = $_SERVER["REQUEST_METHOD"];

if ($method == "GET") {
    $result = mysqli_query($conn, "SELECT * FROM students");
    $students = [];

    while ($row = mysqli_fetch_assoc($result)) {
        $students[] = $row;
    }

    echo json_encode($students);
}

elseif ($method == "POST") {
    $data = json_decode(file_get_contents("php://input"), true);

    $name = $data["name"];
    $email = $data["email"];
    $mobile = $data["mobile"];

    $sql = "INSERT INTO students (name, email, mobile)
            VALUES ('$name', '$email', '$mobile')";

    if (mysqli_query($conn, $sql)) {
        echo json_encode(["message" => "Student Added"]);
    } else {
        echo json_encode(["message" => "Error"]);
    }
```

```php
}

elseif ($method == "PUT") {
    $data = json_decode(file_get_contents("php://input"), true);

    $id = (int)$data["id"];
    $name = mysqli_real_escape_string($conn, $data["name"]);
    $email = mysqli_real_escape_string($conn, $data["email"]);
    $mobile = mysqli_real_escape_string($conn, $data["mobile"]);

    $sql = "UPDATE students SET name='$name',
            email='$email', mobile='$mobile' WHERE id=$id";

    mysqli_query($conn, $sql);

    echo json_encode(["message" => "Student Updated"]);
}

elseif ($method == "DELETE") {
    $id = (int)$_GET["id"];

    mysqli_query($conn, "DELETE FROM students WHERE id=$id");

    echo json_encode(["message" => "Student Deleted"]);
}
?>
```

Output:

Student Registration Form

Name | Email | Mobile | Register

Student Updated

Student List

ID	Name	Email	Mobile	Action
2	diya	diya1337@gmail.com	7016945533	Update Delete
3	yana	vaidehigangani0003@gmail.com	9313731515	Update Delete
4	karan	karan1007@gmail.com	7016945555	Update Delete
6	Vaidehi Gangani	vaidehigangani1367@gmail.com	9313731616	Update Delete
7	Rahul	rahul123@gmail.com	9198765432	Update Delete